



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
CIVIL/AGRICULTURAL & BIOSYSTEMS ENGINEERING

ECV 201: SURVEYING I

1ST SEMESTER 2020/2021

TIME: 2 Hours

INSTRUCTIONS:

Attempt question **ONE** and any other **TWO** questions

QUESTION ONE

(30 Marks)

a) Define the following terms as used in surveying

(6 Marks)

- i. Chaining
- ii. Ranging
- iii. Calibration

b) Reciprocal levelling between two points Y and Z 730m apart on opposite sides of a river gave the following results:

Instrument at	Height of Instrument (m)	Staff at	Staff reading (m)
Y	1.463	Z	1.688
Z	1.436	Y	0.991

Determine the difference in level between Y and Z and the amount of any collimation error in the instrument.

(4 Marks)

c) Describe the phase difference method used in Electromagnetic distance measurement instruments

(4 Marks)

d) The following levels were taken with a metric staff on a series of pegs at 100-m intervals along the line of a proposed trench.

BS	IS	FS	R/F	Remarks
2.10				TBM 28.75 m
	2.85	-0.75		Peg A
1.80		3.51		Peg B
	1.58	+0.22		Peg C
	2.24	-0.66		Peg D
1.68		2.94		Peg E
	2.27			
	3.06			
		3.81		TBM 24.07 m

If the trench is to be excavated from peg A commencing at a formation level of 26.5 m and falling to peg E at a grade of 1 in 200, calculate the height of the sight rails in metres at A, B, C, D and E, if a 3-m boning rod is to be used. (4 Marks)

e) The height of an EDM set up at M is 1.495 m. The height of a reflector set up at P is 1.30 m. The height of the theodolite at M used to measure the vertical angle is 1.615 m. The height of the target at P on which the vertical sight is taken is 1.385 m. the slope distance after meteorological corrections is 1650.452 m. The measured vertical angle is $+3^{\circ}02'32''$. What is the horizontal distance between M and P. (3 Marks)

a) A slope distance of 165.360 m (corrected for meteorological conditions) was measured from A to B, whose elevations were 447.401 and 445.389 m above datum, respectively. Find the horizontal length of line AB if the heights of the EDM instrument and reflector were 1.417 and 1.615 m above their respective stations. (3 Marks)

b) Describe the ranging methods employed in surveying (6 Marks)

(20 Marks)

QUESTION TWO

a) Describe the principle of differential leveling (5 Marks)

b) The frequencies used by an EDM are 15 MHz and 150 kHz. Taking the velocity of light as 299,793 km/s and a measure distance of 346.73 m, show the

computational processes necessary to obtain this distance, clearly illustrating the phase difference technique.

(8 Marks)

c) Describe the Peg Test in leveling

(7 marks)

QUESTION THREE

(20 Marks)

a) Discuss the corrections applied to an electromagnetic distance measurement instrument

(5 Marks)

b) A red laser light ($\lambda = 632.8$ nanometers) operates at a temperature of 301.15 Kelvin, barometric pressure of 710 torr, and a vapour pressure of 30 torr. Given the Barrell and sears (1939) formula for the group index of refraction (n_g) and the ambient refractive index n_a . Considering the speed of light in a vacuum to be 299792.5km/s and a heat expansion coefficient of air 0.00367

(7 Marks)

$$(n_g - 1)10^7 = 2876.04 + \frac{3(16.288)}{\lambda^2} + \frac{5(0.136)}{\lambda^4}$$

$$N_a = \left[\frac{n_g - 1}{1 + \alpha t} \frac{P}{760} - \frac{5.5(10)^{-8}}{1 + \alpha t} e \right] 10^6$$

i. What is the refractive index?

ii. What is the velocity through air?

iii. What is the modulated wavelength if the modulating frequency is 44MHz?

c) Outline the procedure of measuring distance along the ground by use of a tape (8 marks)

QUESTION FOUR

(20 Marks)

a) Describe the various instruments used for the determination of the length of line

(7 Marks)

b) The following readings were taken with a level and a 4.25-m staff: 0.683, 1.109, 1.838, 3.398 [3.877 and 0.451] CP, 1.405, 1.896, 2.676 BM (102.120 AOD), 3.478 [4.039 and 1.835] CP, 0.649, 1.707, 3.722.

- i. Draw up a level book and reduce the levels by the R-and-F and height of collimation methods. (10 Marks)
 - ii. What error would occur in the final level if the staff had been wrongly extended and a plain gap of over 12 mm occurred at the 1.52-m section joint? (3 Marks)
- (20 Marks)

QUESTION FIVE

- a) Describe the pulse method used in Electromagnetic distance measurement instruments. (5 Marks)
- b) Explain the different types of tapes used for distance measurement. (6 Marks)
- c) Although modern EDM equipment is exceptionally well constructed, the effects of age and general wear and tear may result to some errors. Describe these errors and their calibration procedures. (9 Marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 201: SURVEYING I

DATE: Monday, 27th January 2020

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

Attempt questions ONE and any other TWO questions.

QUESTION ONE (30 MARKS)

(a) Define the following terms as used in surveying:

(6 Marks)

(i) Chaining

(ii) Ranging

(iii) Calibration

measuring distance on surface of the earth either directly with the chain or tape and it is of high accuracy over shorter distance

(b) Reciprocal levelling between two points Y and Z 730m apart on opposite sides of a river gave the following results:

Instrument at	Height of instrument (m)	Staff at	Staff reading (m)
Y	1.463	Z	1.688
Z	1.436	Y	0.991

Determine the difference in level between Y and Z and the amount of any collimation error in the instrument.

(4 Marks)

(c) Describe the phase difference method used in Electromagnetic distance measurement instruments.

(4 Marks)

- (d) The following levels were taken with a metric staff on a series of pegs at 100-m intervals along the line of a proposed trench. (4 Marks)

<i>BS</i>	<i>IS</i>	<i>FS</i>	<i>Remarks</i>
2.10			TBM 28.75 m
	2.85		Peg A
1.80		3.51	Peg B
	1.58		Peg C
	2.24		Peg D
1.68		2.94	Peg E
	2.27		
	3.06		
		3.81	TBM 24.07m

If the trench is to be excavated from peg A commencing at a formation level of 26.5 m and falling to peg E at a grade of 1 in 200, calculate the height of the sight rails in metres at A, B, C, D and E, if a 3-m boning rod is to be used. (4 Marks)

- (e) The height of an EDM set up at M is 1.495m. The height of a reflector set up at P is 1.30m. The height of the theodolite at M used to measure the vertical angle is 1.615m. The height of the target at P on which the vertical sight is taken is 1.385m. The slope distance after meteorological corrections is 1650.452m. The measured vertical angle is $+3^{\circ}02'32''$. What is the horizontal distance between M and P? (3 Marks)

- (f) A slope distance of 165.360m (corrected for meteorological conditions) was measured from A to B, whose elevations were 447.401 and 445.389m above datum, respectively. Find the horizontal length of line AB if the heights of the EDM instrument and reflector were 1.417 and 1.615m above their respective stations. (3 Marks)

- (g) Describe the ranging methods employed in surveying. (6 Marks)

QUESTION TWO (20 MARKS)

- (a) Describe the principle of differential leveling. (5 Marks)
- (b) The frequencies used by an EDM are 15 MHz and 150 kHz. Taking the velocity of light as 299,793 km/s and a measure distance of 346.73m, show the computational processes necessary to obtain this distance, clearly illustrating the phase difference technique. (8 Marks)
- (c) Describe the Peg Test in leveling. (7 Marks)

QUESTION THREE (20 MARKS)

- (a) Discuss the corrections applied to an electromagnetic distance measurement instrument. (5 Marks)
- (b) A red laser light ($\lambda = 632.8$ nanometers) operates at a temperature of 301.15 Kelvin, barometric pressure of 710 torr, and a vapour pressure of 30 torr. Given the Barrell and sears (1939) formula for the group index of refraction (n_g) and the ambient refractive index n_a . Considering the speed of light in a vacuum to be 299792.5 km/s and a heat expansion coefficient of air 0.00367. (7 Marks)

$$(n_g - 1)10^7 = 2876.04 + \frac{3(16.288)}{\lambda^2} + \frac{5(0.136)}{\lambda^4}$$

$$N_a = \left[\frac{n_g - 1}{1 + \alpha t} \frac{P}{760} - \frac{5.5(10)^{-8}}{1 + \alpha t} e \right] 10^6$$

- (i) What is the refractive index?
- (ii) What is the velocity through air?
- (iii) What is the modulated wavelength if the modulating frequency is 44MHz?
- (c) Outline the procedure of measuring distance along the ground by use of a tape. (8 Marks)

QUESTION FOUR (20 MARKS)

- (a) Describe the various instruments used for the determination of the length of line. (7 Marks)
- (b) Reduce the given booking data below using
- (i) The Rise & Fall

Timbuktu.

- (ii) Height of collimation methods. The reduced level at the TBM is +43.00.
arithmetic checks to support your results. (13 Marks)

Point	BS	IS	FS
TBM	0.771		
A	0.802		1.552
B		2.311	
C	3.580		1.990
D		1.220	
E		3.675	
F	2.408		4.02
G		0.339	
H			0.157

QUESTION FIVE (20 MARKS)

- (a) Describe the pulse method used in Electromagnetic distance measurement instruments. (5 Marks)
- (b) Explain the different types of tapes used for distance measurement. (6 Marks)
- (c) Although modern EDM equipment is exceptionally well constructed, the effects of age and general wear and tear may result to some errors. Describe these errors and their calibration procedures. (9 Marks)

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2018/2019
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (CIVIL ENGINEERING)
ECV 201: SURVEYING I

DATE: MONDAY 11TH MARCH 2019

TIME: 8.00 A.M. – 10.00 A.M.

INSTRUCTIONS

Answer Question ONE and any other TWO questions

QUESTION ONE

(30 Marks)

- a) Describe the phase difference method used in Electromagnetic distance measurement instruments (4 Marks)
- b) The height of an EDM set up at M is 1.495 m. The height of a reflector set up at P is 1.30 m. The height of the theodolite at M used to measure the vertical angle is 1.615 m. The height of the target at P on which the vertical sight is taken is 1.385 m. the slope distance after meteorological corrections is 1650.452 m. The measured vertical angle is $+3^{\circ}02'32''$. What is the horizontal distance between M and P. (3 Marks)

- c) The following staff readings were observed (in the order given) when levelling up a hillside from a TBM 135.2 m ASL. Excepting the staff position immediately after the TBM, each staff position was higher than the preceding one.

1.408, 2.728, 1.856, 0.972, 3.789, 2.746, 1.597, 0.405, 3.280, 2.012, 0.625, 4.136, 2.664, 0.994, 3.901, 1.929, 3.478, 1.332

Enter and reduce the readings in level-book form by both the R-and-F and collimation systems. (10 marks)

- d) A slope distance of 165.360 m (corrected for meteorological conditions) was measured from A to B, whose elevations were 447.401 and 445.389 m above datum, respectively. Find the horizontal length of line AB if the heights of the EDM instrument and reflector were 1.417 and 1.615 m above their respective stations. (3 Marks)

$$D = 165.3244 \text{ m}$$

- e) Describe the ranging methods employed in surveying (10 Marks)

QUESTION TWO ✓

(20 Marks)

- a) Describe the principle of differential leveling (5 Marks)
- b) The frequencies used by an EDM are 15 MHz and 150 kHz. Taking the velocity of light as 299,793 km/s and a measure distance of 346.73 m, show the computational processes necessary to obtain this distance, clearly illustrating the phase difference technique. (8 Marks)
- c) Describe the Peg Test in leveling (7 marks)

QUESTION THREE

(20 Marks)

- a) Discuss the corrections applied to an electromagnetic distance measurement instrument (5 Marks)
- b) A red laser light ($\lambda = 632.8$ nanometers) operates at a temperature of 301.15 Kelvin, barometric pressure of 710 torr, and a vapour pressure of 30 torr. Given the Barrell and sears (1939) formula for the group index of refraction (n_g) and the ambient refractive index n_a . Considering the speed of light in a vacuum to be 299792.5 km/s and a heat expansion coefficient of air 0.00367 (7 Marks)

$$\begin{aligned} (n_g - 1)10^7 &= 2876.04 + \frac{3(16.288)}{\lambda^2} + \frac{5(0.136)}{\lambda^4} \\ N_a &= \left[\frac{n_g - 1}{1 + \alpha t} \frac{P}{760} - \frac{5.5(10)^{-8}}{1 + \alpha t} e \right] 10^6 \end{aligned}$$

- i. What is the refractive index
- ii. What is the velocity through air?
- iii. What is the modulated wavelength if the modulating frequency is 44 MHz?
- c) Outline the procedure of measuring distance along the ground by use of a tape (8 marks)

QUESTION FOUR ✓

(20 Marks)

- a) Describe the various instruments used for the determination of the length of line (7 Marks)
- b) Reduce the given booking data below using
- i. The Rise & Fall and
- ii. Height of collimation methods. The reduced level at the TBM is +43.00 Use arithmetic checks to support your results (13 Marks)

Point	BS	IS	FS
TBM	0.771		
A	0.802		1.552
B		2.311	
C	3.580		1.990
D		1.220	
E		3.675	
F	2.408		4.02
G		0.339	
H			0.157

QUESTION FIVE

(20 Marks)

- Describe the pulse method used in Electromagnetic distance measurement instruments
(5 Marks)
- Explain the different types of tapes used for distance measurement
(6 Marks)
- Although modern EDM equipment is exceptionally well constructed, the effects of age and general wear and tear may result to some errors. Describe these errors and their calibration procedures
(9 Marks)

KENYATTA UNIVERSITY

SURVEYING 1

TIME: 1 HOUR CAT I

Instructions: Attempt all question

1. Define the following terms as used in surveying (10 Marks)

- i. Horizontal control
- ii. Vertical control
- iii. Route surveys
- iv. Layout survey
- v. Preliminary survey

2. Describe the two (2) ranging techniques used in surveying (5 Marks)

3. A slope distance of 204.834 m was determined by EDM, the instrument and prism having been set up 1.645 m and 1.873 m respectively above their stations. The vertical angle between the stations was later measured as $2^{\circ} 17' 40''$, the instrument station being the higher of the two. Calculate the horizontal length of the line. (4 marks)

4. A red laser light ($\lambda = 0.6328$ micrometers) operates at a temperature of 34°C , barometric pressure of 710 torr and a relative humidity of 56%. The partial water vapor pressure, e , can be computed with sufficient accuracy for normal operating conditions as

$$e = E \cdot h\% \quad \text{Where } E = 10^{\left[\frac{7.5t}{237.3+t} + 0.7858\right]} \text{ and } h \text{ is the relative humidity in percent.}$$

- i. Compute the refractive index
- ii. What is the velocity through air?
- iii. What is the modulated wavelength if the modulating frequency is 34MHz?

Given the Barrell and sears (1939) formula for the group index of refraction (n_g) and the ambient refractive index n_a . The velocity of light in a vacuum is 299792.5km/s and a heat expansion coefficient of air 0.00367. (7 marks)

$$(n_g - 1)10^7 = 2876.04 + \frac{3(16.288)}{\lambda^2} + \frac{5(0.136)}{\lambda^4},$$

$$N_a = \left[\frac{n_g - 1}{1 + \alpha t} \frac{P}{760} - \frac{5.5(10)^{-8}}{1 + \alpha t} e \right] 10^6$$

5. A slope distance of 1650.452 m (corrected for meteorological conditions) was measured from M to P, whose elevations were 447.401 and 445.389 m above datum, respectively. The height of an EDM set up at M is 1.495 m. The height of a reflector set up at P is 1.295 m. The elevation of the theodolite at M used to measure the vertical angle is 449.116 m above datum. The elevation of the target at P on which the vertical sight is taken is 446.779 m above datum. The slope distance after meteorological corrections is 1650.452 m. The measured vertical angle is $-3^{\circ} 02' 23''$. What is the horizontal distance between M and P. (4 marks)

(10 marks)

1. Define the following terms as used in surveying
 - i. Horizontal control
 - ii. Orthometric height
 - iii. Ellipsoidal height
 - iv. Layout survey
 - v. Preliminary survey
2. Describe the two (2) ranging techniques used in surveying (5 marks)
3. A red laser light ($\lambda = 0.6328$ micrometers) operates at a temperature of 34°C , barometric pressure of 710 torr and a relative humidity of 56%, The partial water vapor pressure, e , can be computed with sufficient accuracy for normal operating conditions as

$$e = E.h\% \quad \text{Where } E = 10^{\left[\frac{7.5t}{237.3+t} + 0.7858\right]} \text{ and } h \text{ is the relative.}$$

- i. Compute the refractive index
- ii. What is the velocity though air?
- iii. What is the modulated wavelength if the modulating frequency is 34MHz?

Given the Barrell and sears (1939) formula for the group index of refraction (n_g) and the ambient refractive index n_a . The velocity of light in a vacuum is 299792.5km/s and a heat expansion coefficient of air 0.00367.
(7 marks)

$$(n_g - 1)10^7 = 2876.04 + \frac{3(16.288)}{\lambda^2} + \frac{5(0.136)}{\lambda^4},$$

$$N_a = \left[\frac{n_g - 1}{1 + \alpha t} \frac{P}{760} - \frac{5.5(10)^{-8}}{1 + \alpha t} e \right] 10^6$$

4. Describe four applications of leveling in Civil Engineering. (4 marks)
5. To obtain the zero error of a particular EDM instrument, a base line AD is split into three sections AB, BC and CD and measured in the following combinations:

$$AB = 20.512, AC = 63.192, AD = 153.303$$

$$BC = 42.690, BD = 132.803, CD = 90.1201$$

Using all possible combinations, compute the zero error. (4 marks)

Instructions: Attempt all questions:

1. Define following terminologies as used in surveying

(8 marks)

- Peg test
- Reciprocal leveling
- Indirect contouring
- Elevation

2. Outline the properties of contour lines

(7 marks)

3. A sewer is to be laid at a uniform gradient of 1 in 200, between two points X and Y, 240 m apart. The reduced level of the invert at the outfall X is 150.82. In order to fix sight rails at X and Y, readings are taken with a level in the following order:

(8 marks)

Reading	Staff station
BS 0.81	TBM (near X), RL 153.81
IS 'a'	Top of sight rail at X
IS 1.07	Peg at X
FS 0.55	CP between X and Y
BS 2.15	CP between X and Y
IS 'b'	Top of sight rail at Y
FS 1.88	Peg at Y

- Draw up a level book and find the reduced levels of the pegs.
 - If a boning rod of length 3m is to be used, find the level readings *a* and *b*.
 - Find the height of the sight rails above the pegs at X and Y.
4. The following staff readings were observed (in the order given) when levelling up a hillside from a TBM 135.2 m ASL. Excepting the staff position immediately after the TBM, each staff position was higher than the preceding one.
- 1.408, 2.728, 1.856, 0.972, 3.789, 2.746, 1.597, 0.405, 3.280, 2.012, 0.625, 4.136, 2.664, 0.994, 3.901, 1.929, 3.478, 1.332
- Enter and reduce the readings in level-book form by the R-and-F systems.

(8 marks)

Instructions: Attempt all questions:

1. Define following terminologies as used in surveying (8 marks)
 - a. Peg test *line of sight*
 - b. Reciprocal leveling *line of sight*
 - c. Indirect contouring *traverse*
 - d. Precise leveling
2. Outline the properties of contour lines (7 marks)
3. In leveling across a river, reciprocal leveling observations gave the following results for staffs held vertically at X and Y from level stations A and B on each bank respectively:

Staff reading of X from A = 1.753m *a₁*

Staff reading of X from B = 2.080 m *a₂*

Staff reading of Y from A = 2.550 m *b₁*

Staff reading of Y from B = 2.895 m *b₂*

*If the reduced level of X is 90.37 ASL, Obtain that of Y. (2 marks)

4. State five uses of contours in Engineering (5 marks)

5. A sewer is to be laid at a uniform gradient of 1 in 200, between two points X and Y, 240 m apart. The reduced level of the invert at the outfall X is 150.82. In order to fix sight rails at X and Y, readings are taken with a level in the following order: (8 marks)

Reading	Staff station
BS 0.81	TBM (near X), RL 153.81
IS 'a'	Top of sight rail at X
IS 1.07	Peg at X
FS 0.55	CP between X and Y
BS 2.15	CP between X and Y
IS 'b'	Top of sight rail at Y
FS 1.88	Peg at Y

- i. Draw up a level book and find the reduced levels of the pegs.
- ii. If a boning rod of length 3m is to be used, find the level readings a and b.
- iii. Find the height of the sight rails above the pegs at X and Y.